# Brief Introduction of Meteorological Data Modeling

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Data Mining
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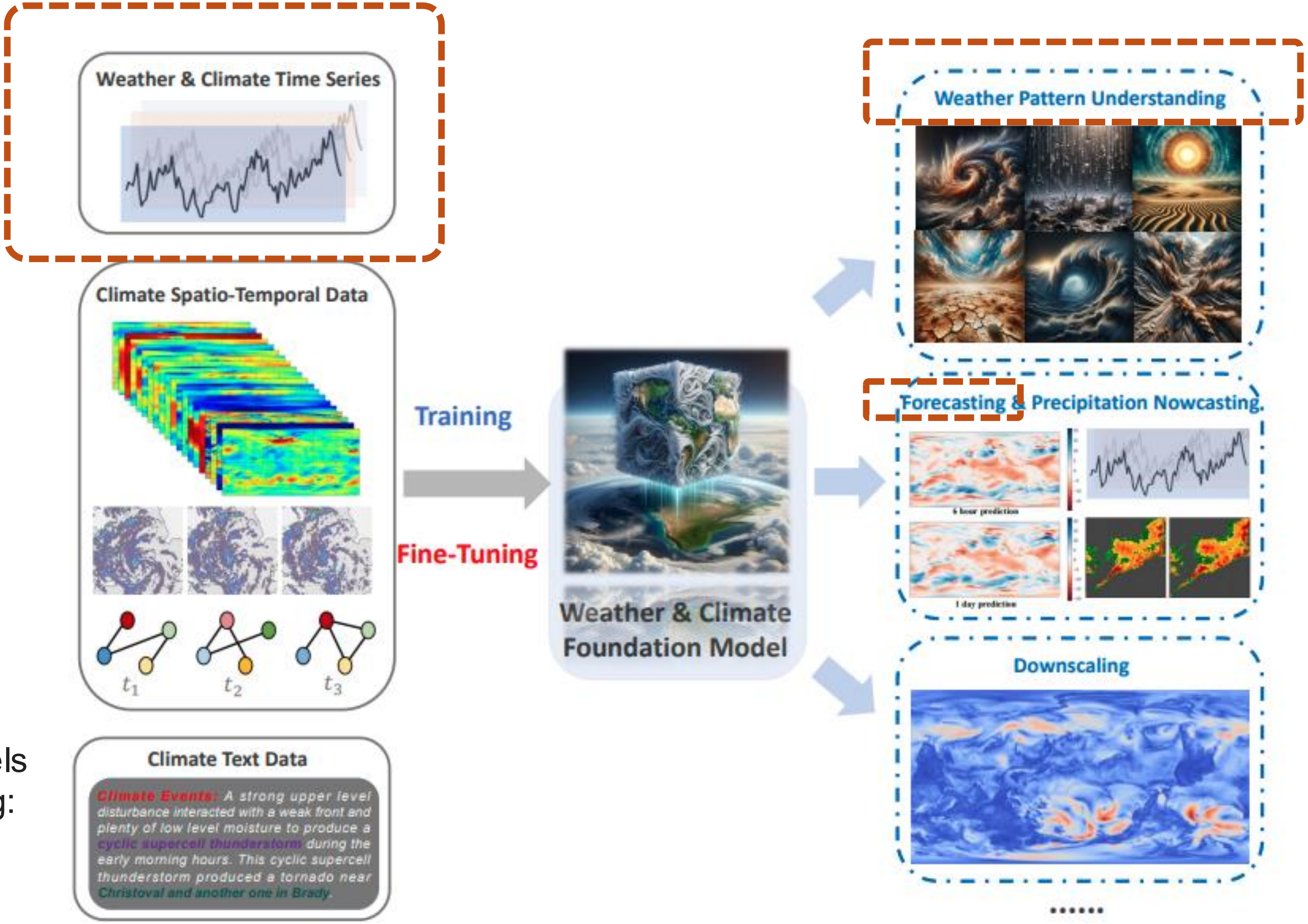
# Announcement

- For the final project submission on Kaggle, please make sure each group uses one account for submission.

- Regular course teaching
  - May 28: Time Series Analysis
  - June 4: Reinforcement learning and advanced frameworks
  - The third hour will be an asynchronous class on May 28 and June 4

- Details of final project presentation will be released on May 28

# Weather and Climate Data




Chen, Shengchao, et al. "Foundation models for weather and climate data understanding: A comprehensive survey." *arXiv preprint arXiv:2312.03014* (2023).

# Weather and Climate Modeling

- Task-Specific Models for Time Series Data
  - Transformer-based
  - CNN / RNN / GNN-based
  - Diffusion models
  - …

TABLE 3: List of representative models under mainstream applications for weather and climate data. Each column represents, in turn, data type, model category, method, scope, specific task/domain, base model, institution, and year of publication, respectively, noting that the base model is dominated by the primary starter module. More details available in Section. 7.

| Data | Category | Method | Scope | Task/Domain | Base | Institute | Year |
|---|---|---|---|---|---|---|---|
| Time Series | Large Foundation Models | FourCastNet [136] | Task-Specific | Forecasting | RNN | Nvidia | 2022 |
| | | GraphCast [137] | Task-Specific | Forecasting | GNN | Google | 2022 |
| | | PanGu-Weather [63] | Task-Specific | Forecasting | Transformer | HUAWEI | 2023 |
| | | FengWu [138] | Task-Specific | Forecasting | Transformer | SHAI | 2023 |
| | | FuXi [139] | Task-Specific | Forecasting | Transformer | FuDan | 2023 |
| | | W-MAE [64] | Task-Specific | Forecasting | Transformer | UESTC | 2023 |
| | | ClimaX [25] | General | Forecasting/Downscaling | Transformer | UCLA | 2023 |
| | Task-Specific Models | PGnet | Task-Specific | Forecasting | GAN | CAAC | 2021 |
| | | Graphino [140] | Task-Specific | Forecasting | GNN | TUD | 2021 |
| | | TemperatureGAN [141] | Task-Specific | Forecasting | GAN | FuDan | 2021 |
| | | Keisler GNN [142] | Task-Specific | Forecasting | GNN | NONE | 2022 |
| | | GE-STDGN [143] | Task-Specific | Forecasting | RNN, GNN | SEU | 2022 |
| | | HiSTGNN [144] | Task-Specific | Forecasting | GNN | SWJTU | 2022 |
| | | DWFH [145] | Task-Specific | Forecasting | RNN | UHK | 2023 |
| | | SWINRDW [146] | Task-Specific | Forecasting | RNN, Diffusion Models | Alibaba | 2023 |
| | | SWINVRNN [147] | Task-Specific | Forecasting | RNN | Alibaba | 2023 |
| | | PoET [148] | Task-Specific | Forecasting | Transformer | ECMWF | 2023 |
| | | OceanFourCastNet [149] | Task-Specific | Forecasting | Transformer | MIT | 2023 |
| | | SEEDS [150] | Task-Specific | Forecasting | Diffusion Model | Google | 2023 |
| | | Dyfussion [151] | Task-Specific | Forecasting | Diffusion Model | UCSD | 2023 |
| | | DITTO [152] | Task-Specific | Forecasting | Diffusion Model | TelAviv | 2023 |
| | | TeleViT [153] | Task-Specific | Forecasting | Transformer | NOA | 2023 |
| | | MetePFL [24] | Task-Specific | Forecasting | Transformer | UTS | 2023 |
| | | FedWing [154] | Task-Specific | Forecasting | Transformer | UTS | 2023 |
| | | FuXi-Extreme [155] | Task-Specific | Forecasting | Transformer | FuDan | 2023 |
| | | Meshfreeflownet [156] | Task-Specific | Downscaling | CNN | UCB | 2020 |
| | | Cdanet [157] | Task-Specific | Downscaling | CNN | KAUST | 2022 |
| | | GPCHC [158] | Task-Specific | Downscaling | GAN | Mila | 2022 |
| | | Bayesian AIG-Transformer [159] | Task-Specific | Downscaling | Transformer | UH | 2022 |
| | | Deepesd [160] | Task-Specific | Downscaling | CNN | IFCA | 2022 |
| | | Multi-Varibales HP [161] | Task-Specific | Downscaling | CNN | IFCA | 2023 |
| | | PhysicsDL [162] | Task-Specific | Downscaling | RNN | Mila | 2023 |
| | | Torchclim [163] | Task-Specific | Downscaling | CNN | UNSW | 2023 |
| | | ResDiff [164] | Task-Specific | Downscaling | Diffusion Model | SDU | 2023 |
| | | MetNet [165] | Task-Specific | Precipitation Nowcasting | CNN | Google | 2020 |
| | | Nowformer [166] | Task-Specific | Precipitation Nowcasting | Transformer | KAIST | 2021 |
| | | MPL-GAN [167] | Task-Specific | Precipitation Nowcasting | GAN | JCU | 2021 |
| | | CMGAT [168] | Task-Specific | Precipitation Nowcasting | GNN | NUDT | 2021 |
| | | GDE [169] | Task-Specific | Precipitation Nowcasting | Diffusion Model | UNIBO | 2021 |
| | | DMSF-GAN [84] | Task-Specific | Precipitation Nowcasting | GAN | GBAMWF | 2022 |
| | | PCT-CYCLEGAN [170] | Task-Specific | Precipitation Nowcasting | GAN | KMA | 2022 |
| | | Rainformer [171] | Task-Specific | Precipitation Nowcasting | Transformer | ZUT | 2022 |
| | | EarthFormer [172] | Task-Specific | Precipitation Nowcasting | Transformer | HKUTS | 2022 |
| | | PTCT [173] | Task-Specific | Precipitation Nowcasting | Transformer | SYSU | 2022 |
| | | MM-RNN [174] | Task-Specific | Precipitation Nowcasting | RNN | HIT | 2023 |
| | | STIN [175] | Task-Specific | Precipitation Nowcasting | RNN | UCAS | 2023 |
| | | TempEE [107] | Task-Specific | Precipitation Nowcasting | Transformer | GBAMWF | 2023 |
| | | Preformer [176] | Task-Specific | Precipitation Nowcasting | Transformer | UCAS | 2023 |
| | | PreDiff [177] | Task-Specific | Precipitation Nowcasting | Diffusion Model | HKUTS | 2023 |
| | | MetNet-3 | Task-Specific | Precipitation Nowcasting | CNN | Google | 2023 |
| | | ENSOTR [178] | Task-Specific | Weather Pattern Understanding | Transformer | FJNU | 2021 |
| | | All-seanson CNN [179] | Task-Specific | Weather Pattern Understanding | CNN | CNU | 2021 |
| | | CNN-GRU [180] | Task-Specific | Weather Pattern Understanding | RNN | USQ | 2021 |
| | | ENSO-GTC [181] | Task-Specific | Weather Pattern Understanding | GNN | TJU | 2022 |
| | | DLHF [74] | Task-Specific | Weather Pattern Understanding | CNN | UCB | 2022 |
| | | ARIMA-LSTM [182] | Task-Specific | Weather Pattern Understanding | RNN | NCUWE | 2022 |
| | | ENSO-ConvGRU [183] | Task-Specific | Weather Pattern Understanding | RNN | UIUC | 2023 |
| | | DK-STN [184] | Task-Specific | Weather Pattern Understanding | GNN | JLU | 2023 |
| | | STIEF [66] | Task-Specific | Weather Pattern Understanding | RNN | UCAS | 2023 |
| | | XDL-CN [65] | Task-Specific | Weather Pattern Understanding | CNN | NEU | 2023 |
| | | CUNet [185] | Task-Specific | Bias Correction | CNN | OUC | 2021 |
| | | SVMBC [186] | Task-Specific | Bias Correction | CNN | UTokyo | 2022 |
| | | Birectional GRU [187] | Task-Specific | Bias Correction | RNN | WHU | 2022 |
| | | Bi-LSTM [188] | Task-Specific | Bias Correction | RNN | FuDan | 2022 |
| | | LSTM-TCN [189] | Task-Specific | Bias Correction | RNN | Verisk | 2022 |
| | | BLSTM-AM-GS [190] | Task-Specific | Bias Correction | RNN | CUST | 2022 |
| | | SRDRN [191] | Task-Specific | Bias Correction | CNN | AU | 2022 |
| | | DIcorrector-remapper [192] | Task-Specific | Bias Correction | CNN | WUSTL | 2022 |
| | | UNIT+QM [193] | Task-Specific | Bias Correction | GAN | UOE | 2023 |
| | | Weathergnn [194] | Task-Specific | Bias Correction | GNN | Alibaba | 2023 |

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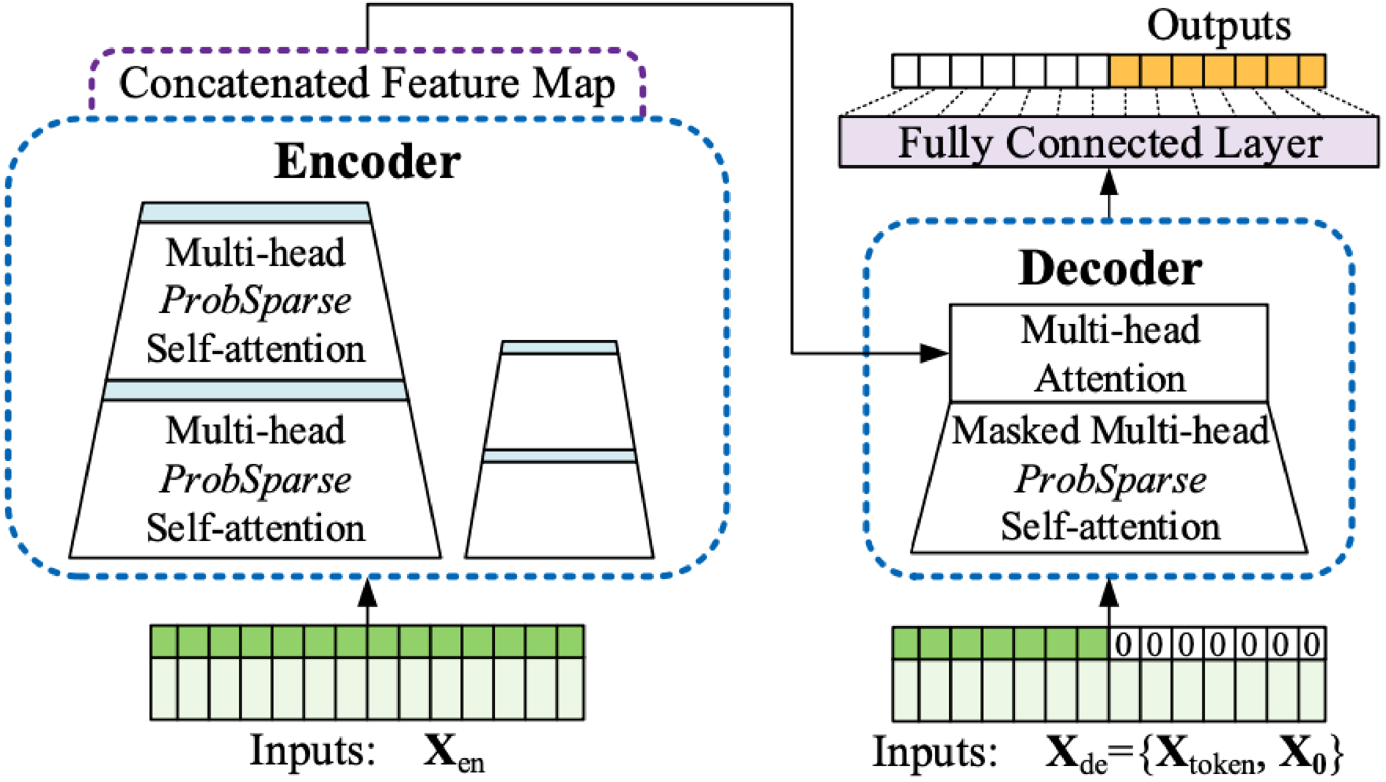
# Transformer-based Models

- Informer [1]
- CrossFormer [2]
- Reformer [3]
- Fedformer [4]
- …

[1] Zhou, Haoyi, et al. "Informer: Beyond efficient transformer for long sequence time-series forecasting." *Proceedings of the AAAI conference on artificial intelligence*. Vol. 35. No. 12. 2021.
[2] Zhang, Yunhao, and Junchi Yan. "Crossformer: Transformer utilizing cross-dimension dependency for multivariate time series forecasting." *The eleventh international conference on learning representations*. 2023.
[3] Kitaev, Nikita, Łukasz Kaiser, and Anselm Levskaya. "Reformer: The efficient transformer." *arXiv preprint arXiv:2001.04451* (2020).
[4] Zhou, Tian, et al. "Fedformer: Frequency enhanced decomposed transformer for long-term series forecasting." *International conference on machine learning*. PMLR, 2022.

# Transformer-based Models

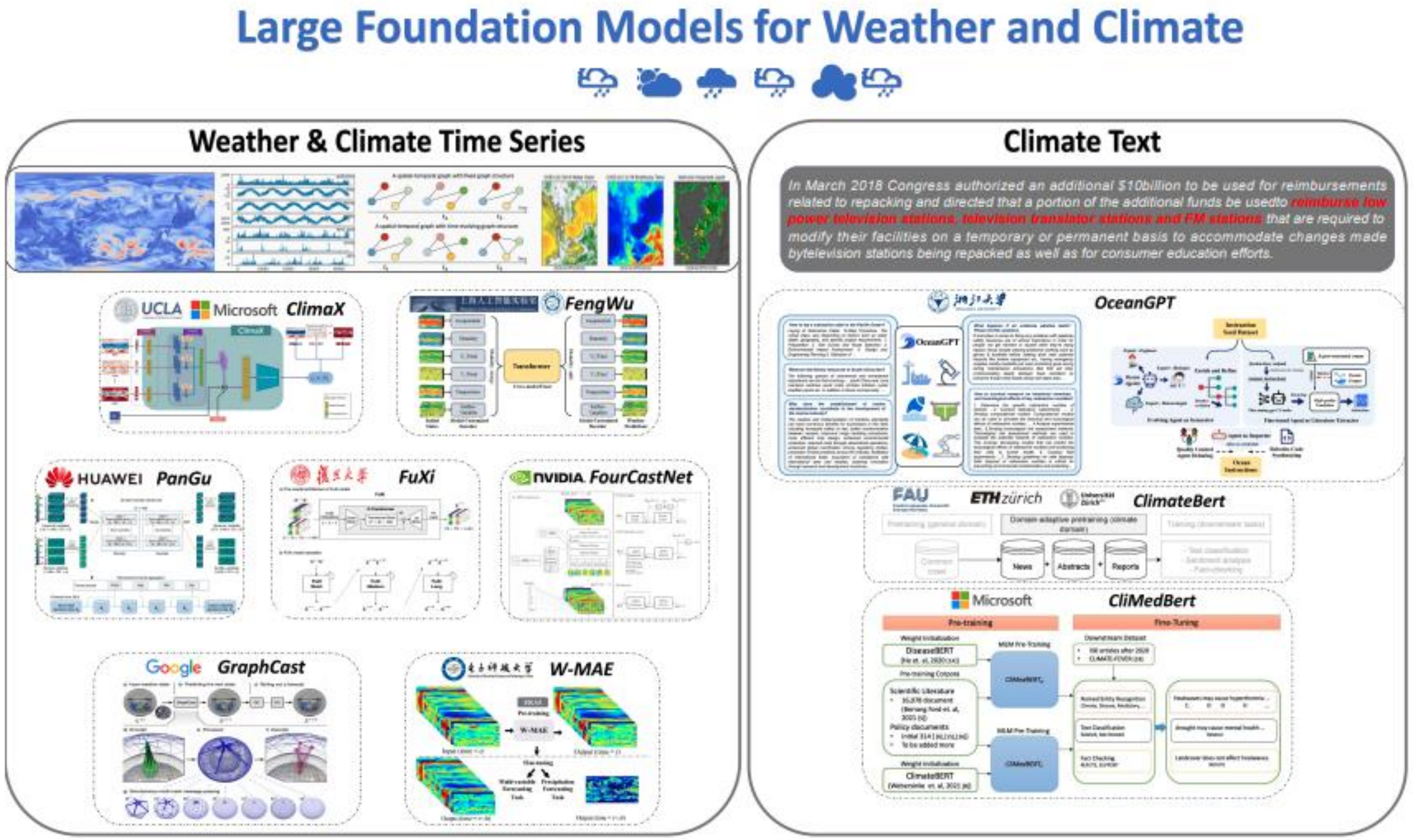
- Informer [1] : Long sequence time-series forecasting (LSTF)
    - Input: $\{\mathbf{x}_1^t, \ldots, \mathbf{x}_{L_x}^t \mid \mathbf{x}_i^t \in \mathbb{R}^{d_x}\}$ at time $t$
    - Output $\{\mathbf{y}_1^t, \ldots, \mathbf{y}_{L_y}^t \mid \mathbf{y}_i^t \in \mathbb{R}^{d_y}\}$





[1] Zhou, Haoyi, et al. "Informer: Beyond efficient transformer for long sequence time-series forecasting." *Proceedings of the AAAI conference on artificial intelligence*. Vol. 35. No. 12. 2021.

# Foundation Models

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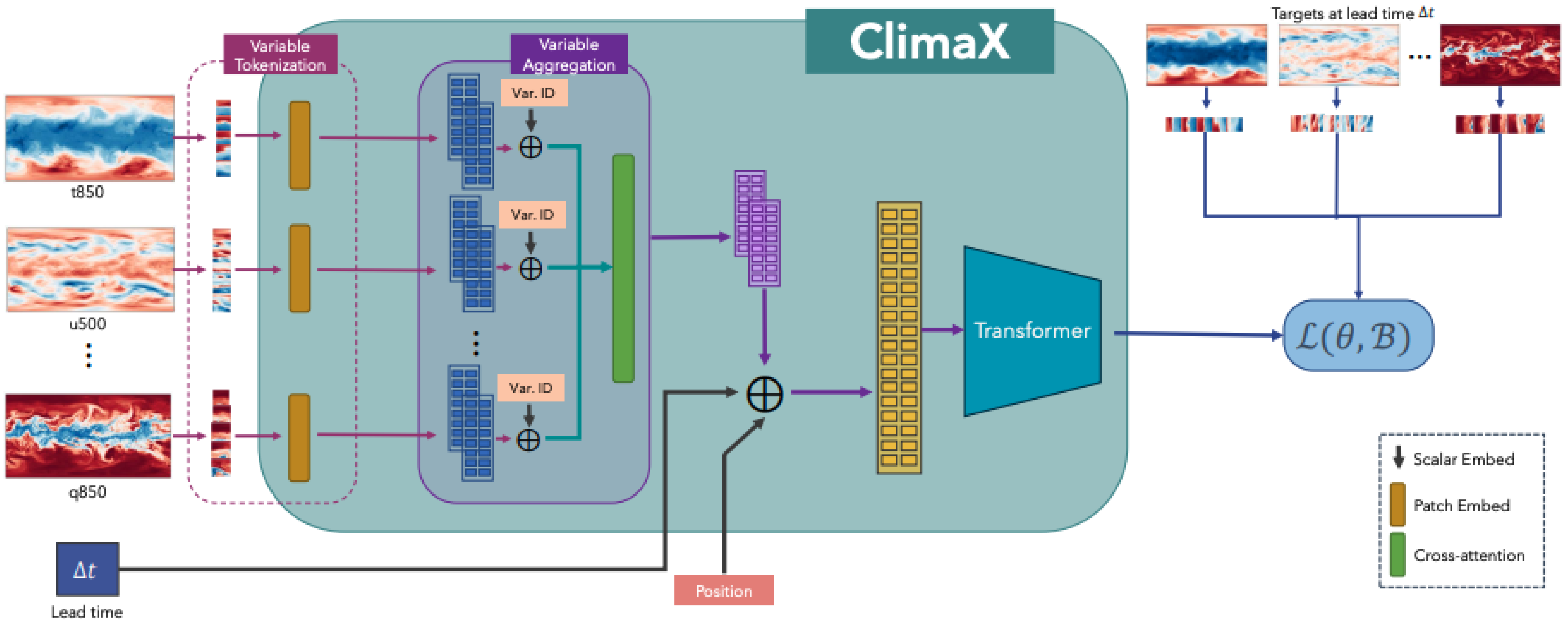
### 8.2 Tools and Models

In this subsection, we collect and compile a rich and usable set of tools and foundation models for modeling weather and climate data.

- **OpenCastKit:** A new global AI weather forecasting project based on FourCastNet and GraphCast. https://github.com/HFAiLab/OpenCastKit
- **GraphCast:** A foundation model for medium-range global weather forecasting. https://github.com/google-deepmind/graphcast
- **FourCastNet:** A foundation model for weather and climate data based on AFNO. https://github.com/NVlabs/FourCastNet
- **PanGu-Weather:** A foundation model for medium-range glocal weather forecasting. https://github.com/198808xc/Pangu-Weather
- **FuXi:** A forecasting system for 15-day global weather forecast. https://github.com/tpys/FuXi
- **W-MAE:** A unsupervised learning global weather forecasting model via Masked Autoencoder. https://github.com/Gufrannn/W-MAE
- **ClimaX:** A versatile climatefoundation model covering forecasting, projection, and downscaling. https://github.com/microsoft/ClimaX
- **OceanGPT:** A large language model for ocean science tasks trained with KnowLM. https://huggingface.co/zjunlp/OceanGPT-7b
- **ClimateBert:** An algorithm that enables to analyze climate-risk disclosures along the four main TCFD categories. https://huggingface.co/climatebert
- **Climate X Quantus:** An XAI toolbox for ML/DL-based climate models. https://github.com/philine-bommer/Climate_X_Quantus

# Weather and Climate Data

- ClimaX [1]
  - Microsoft Github: https://microsoft.github.io/ClimaX/




[1] Nguyen, Tung, et al. "Climax: A foundation model for weather and climate." *arXiv preprint arXiv:2301.10343* (2023)

# Ensemble Methods

**Recap Lecture 07**

- Boosting & Bagging

## Ensemble Methods

- Combine multiple classifiers to improve the performance
- Classic ensemble methods
  - **Bagging**
    - Parallel ensemble learning
    - Each model is trained on a subset of the training data, with all models learned in parallel.
      - E.g., Random Forest
  - **Boosting**
    - Sequential ensemble learning
    - Each new model focuses on the training samples that were misclassified by previous models, and the models are learned sequentially.
      - E.g., AdaBoost, Gradient Boosting *(scalable implementation: XGBoost)*

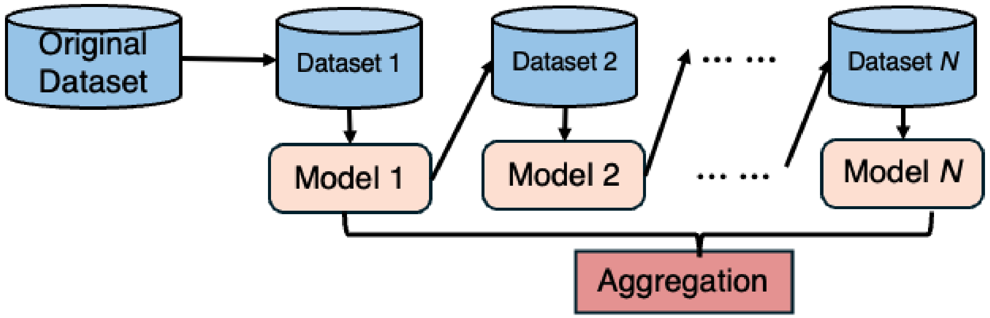
# Ensemble Methods

## Recap Lecture 07

- Boosting

## Boosting

- Each new model focuses on the training samples that were misclassified by previous models, and the models are learned sequentially.
  - Training Process:
    - Train a sequence of models
    - Each model is trained sequentially, with sample weights adjusted based on the <u>errors of previous models</u>
  - Testing Process:
    - Combine the predictions of all models to produce the final output




Data Mining | Spring 2026 Page 23 Lecture 7: Nearest Neighbor Classification and Ensemble Methods

# Ensemble Methods

- Boosting
  - Adaboost [1]
  - LightGBM [2]
  - …

[1] Freund, Yoav, and Robert E. Schapire. "A decision-theoretic generalization of on-line learning and an application to boosting." *Journal of computer and system sciences* 55.1 (1997): 119-139.
[2] Ke, Guolin, et al. "Lightgbm: A highly efficient gradient boosting decision tree." *Advances in neural information processing systems* 30 (2017).

# Data Observation / Preprocessing

- You can first observe the data features and label distribution.

- A validation set separated from the training data is recommended, as it can help prevent your model from overfitting.
  - You can check whether your model is overfitting based on the weekly released private leaderboard.

- Other ideas:
  - Statistics from the past N days, such as mean, standard deviation, etc.
  - The importance of recent versus older data.
  - …

# References


- Chen, Shengchao, et al. "Foundation models for weather and climate data understanding: A comprehensive survey." *arXiv preprint arXiv:2312.03014* (2023).
- Zhou, Haoyi, et al. "Informer: Beyond efficient transformer for long sequence time-series forecasting." *Proceedings of the AAAI conference on artificial intelligence*. Vol. 35. No. 12. 2021.
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- Nguyen, Tung, et al. "Climatelearn: Benchmarking machine learning for weather and climate modeling." *Advances in Neural Information Processing Systems* 36 (2023): 75009-75025.
- Nguyen, Tung, et al. "Climax: A foundation model for weather and climate." *arXiv preprint arXiv:2301.10343* (2023)